Adaptability and the Pivot Penalty in Science and Technology

Impact declines the further researchers move from their prior work

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Northwestern · Cornell · UC Berkeley · NBER · **[[TODO: venue and date]]**

Science is constantly asked to move to new ground

- A pandemic arrives in January and needs answers by March
- Climate change redraws what a dozen fields are for
- Artificial intelligence opens questions nobody was trained on

Every one of these asks working researchers to **work somewhere they have not worked before**.

Two literatures predict opposite answers

Exploration pays.

Novelty, distant search and outsider perspective break problems open. Reaching further should be rewarded.

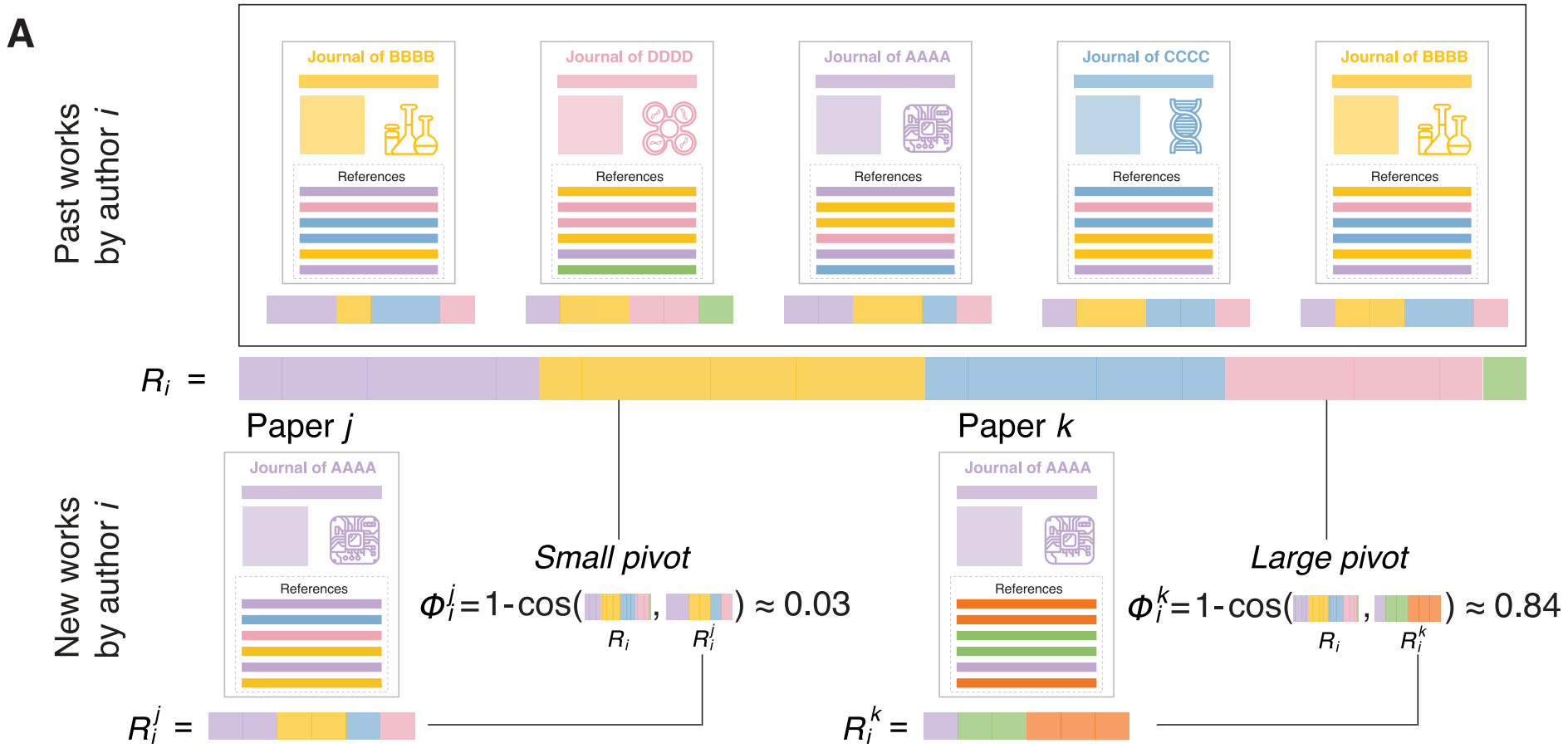
Expertise pays.

Results stand on accumulated facts, theories and tools. Reaching further leaves all of it behind.

Both are well evidenced. **Which one describes a researcher changing direction?**

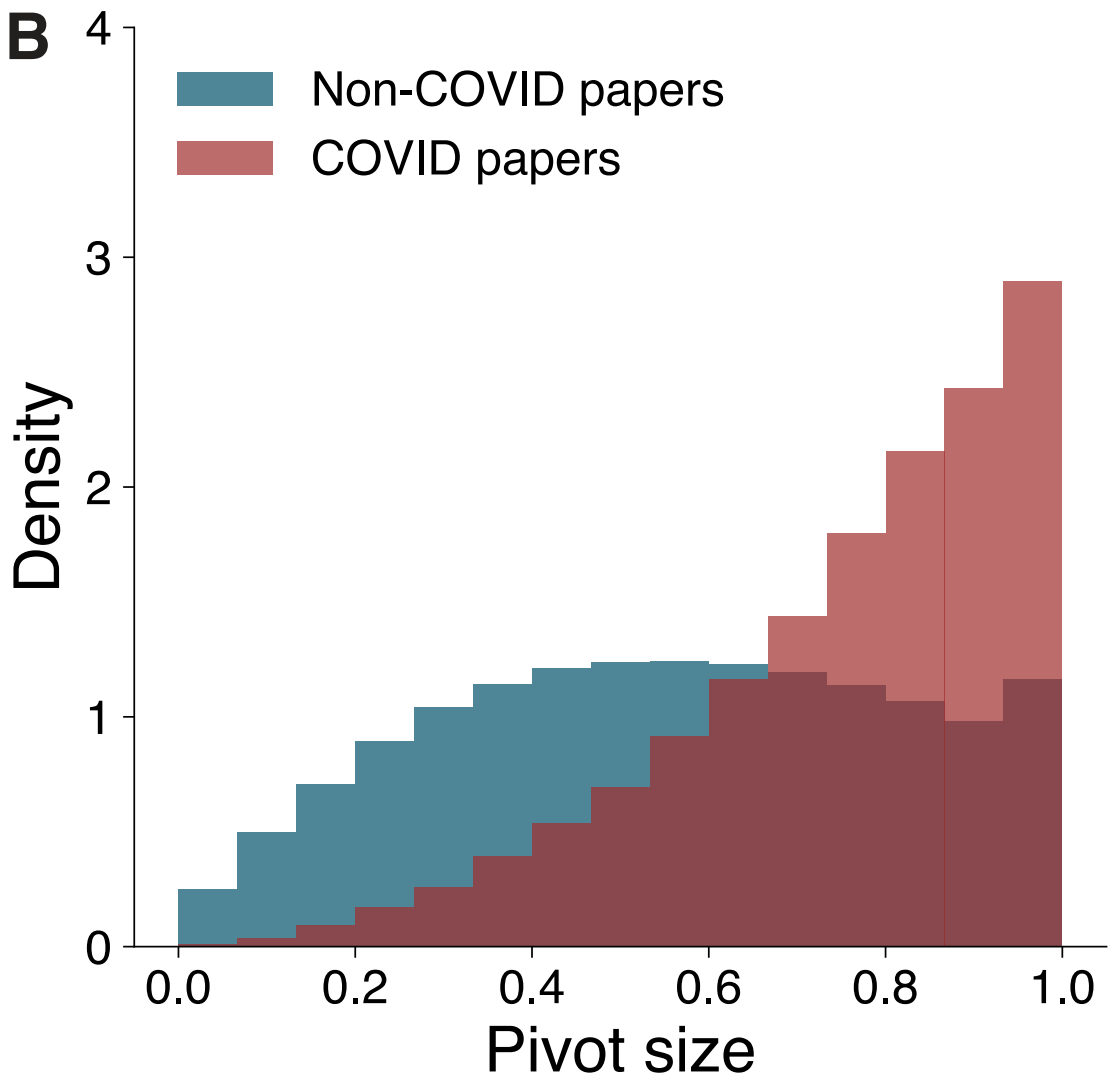
A pivot is the distance from your own past work

between the journals this paper cites and the ones you used to cite



0 = the same journals as before · **1** = entirely new ones · patents use technology classes

Pivoting is common, and its size varies enormously



0.60

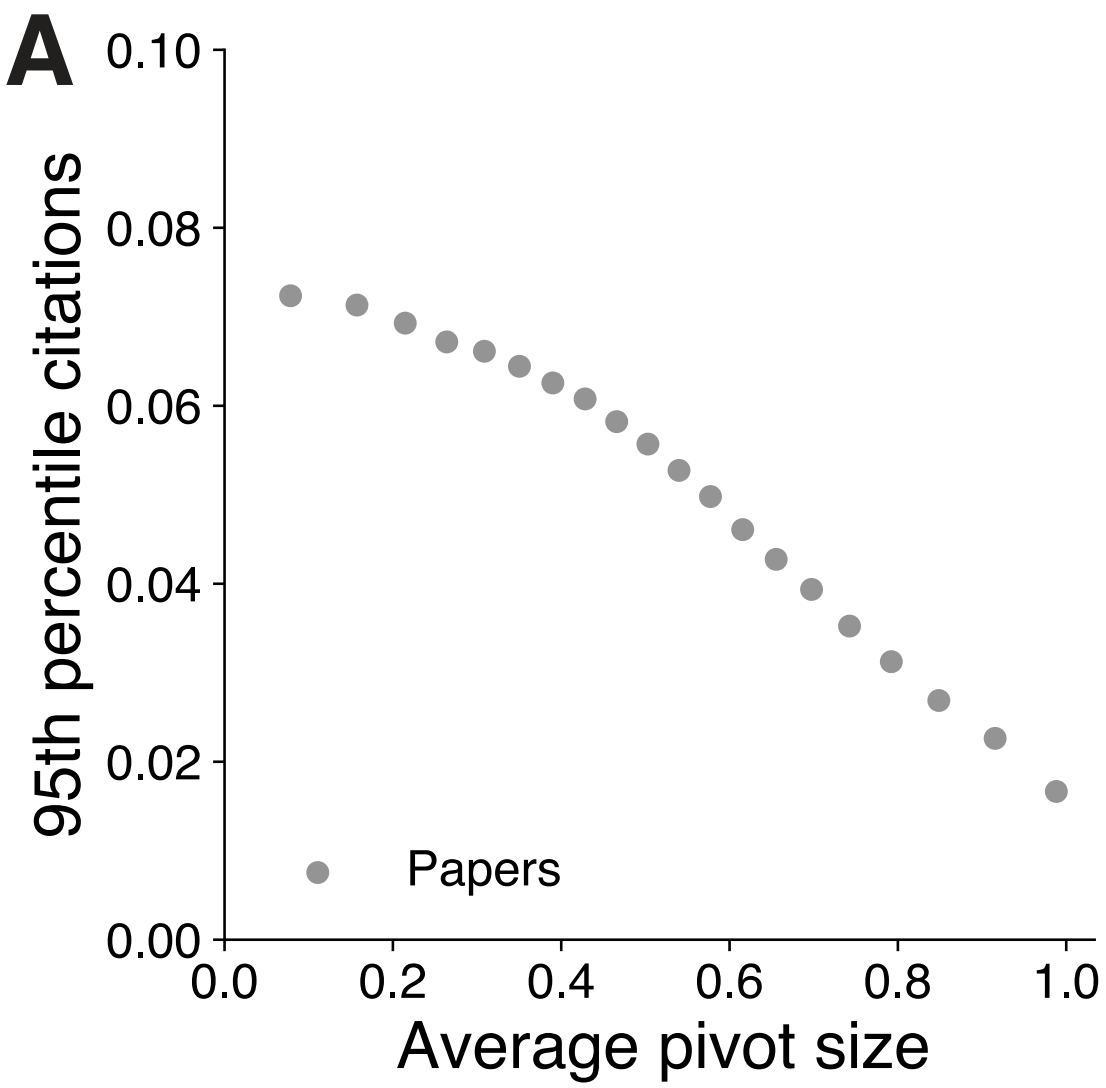
median pivot — ordinary 2020 papers

0.82

median pivot — COVID-19 papers

All 2020 publications indexed by Dimensions

The further researchers move, the less their impact



7.4%

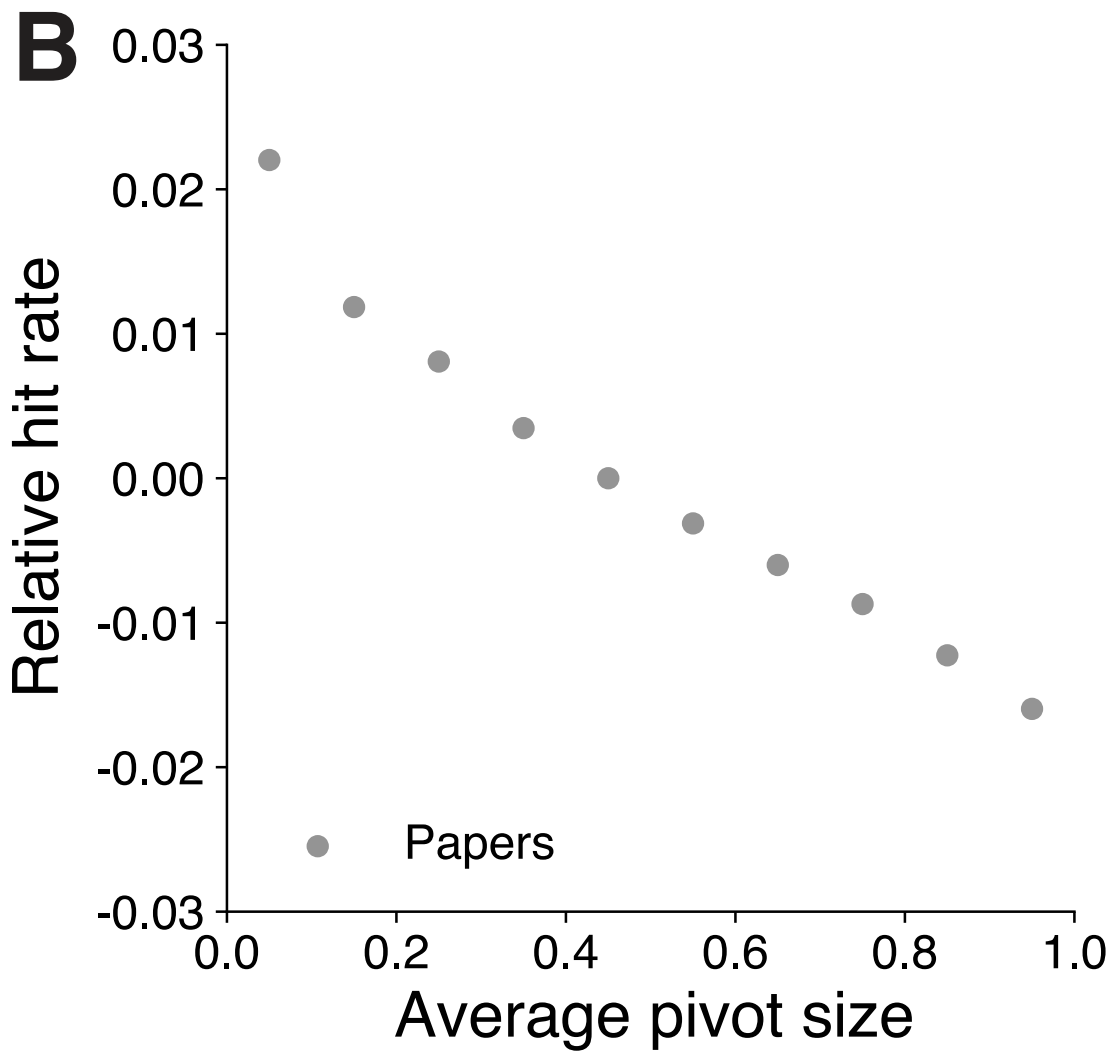
of the smallest pivots are high impact

2.2%

of the largest pivots are

26 million papers · 1970–2015 · 154 fields · high impact
= top 5% of citations in field and year

Researchers do worse on their own bigger pivots



+2.1pp

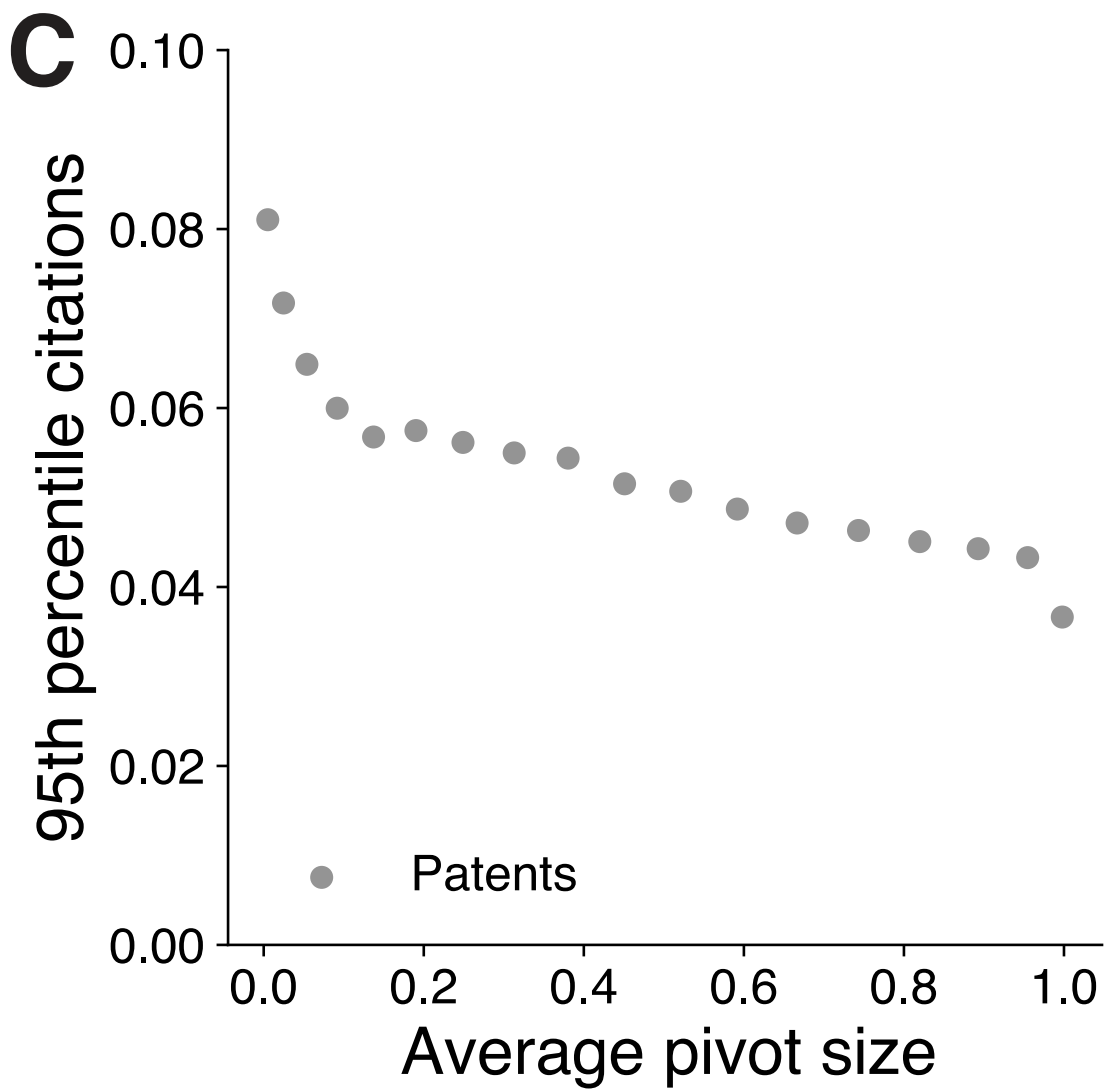
on their own smallest pivots

−1.8pp

on their own largest pivots

> Not a story about **who** pivots.

Inventors pay the same penalty



8.0%

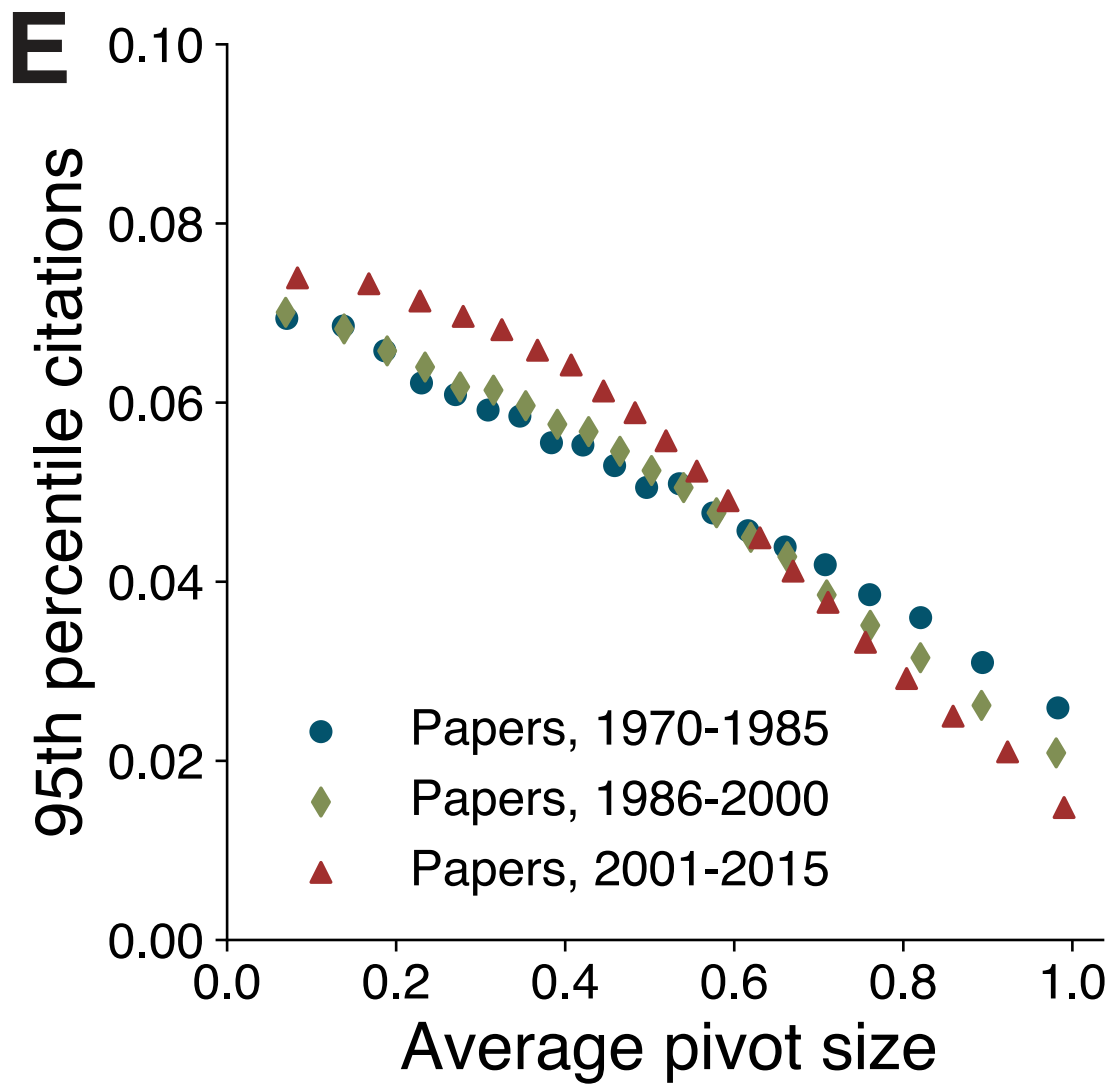
of the smallest pivots are high impact

3.8%

of the largest pivots are

1.8 million US patents · 1980–2015 · 127 technology classes

The penalty has steepened over five decades



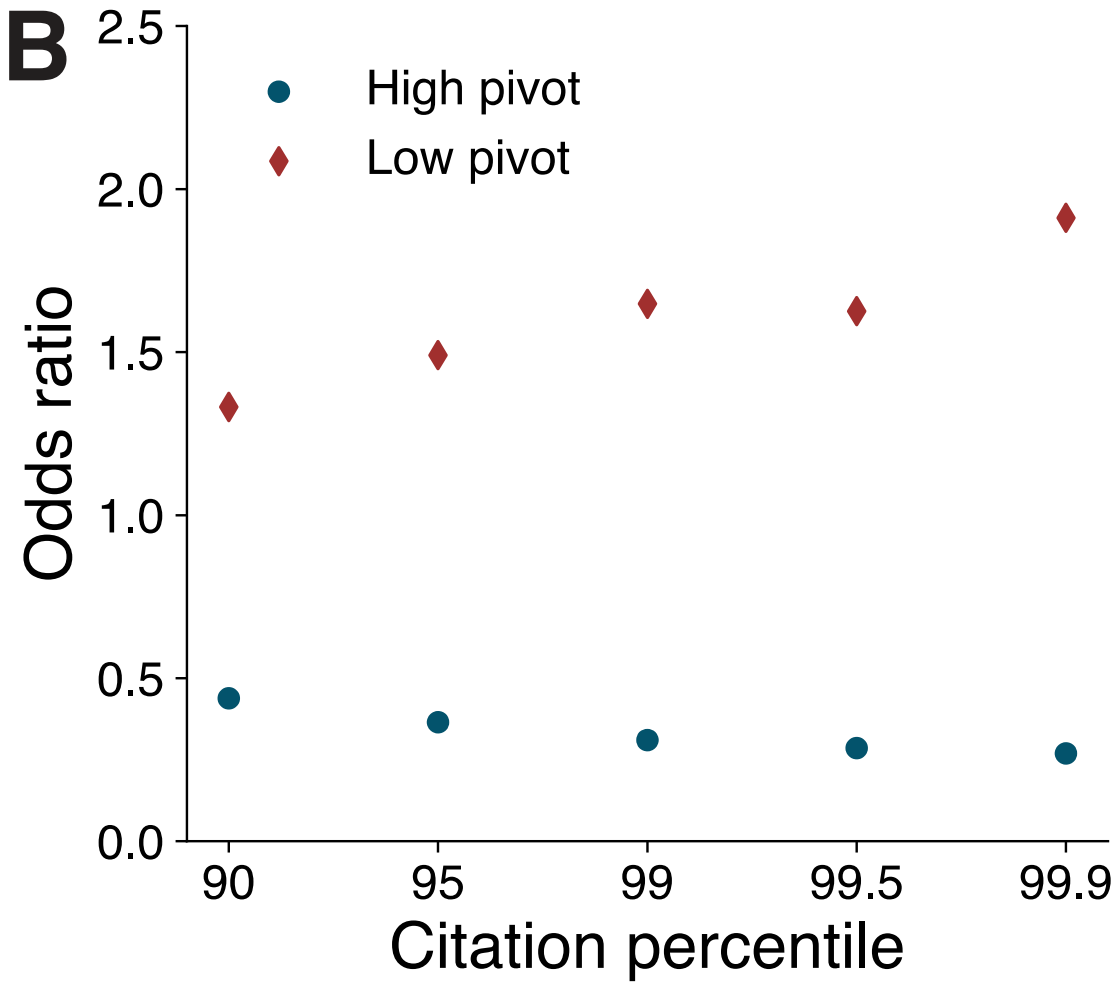
88%

of 154 fields show the penalty steepening over time

Adaptability is getting **harder, not easier.**

Large pivots are not high-risk, high-reward

they lose the top of the distribution too



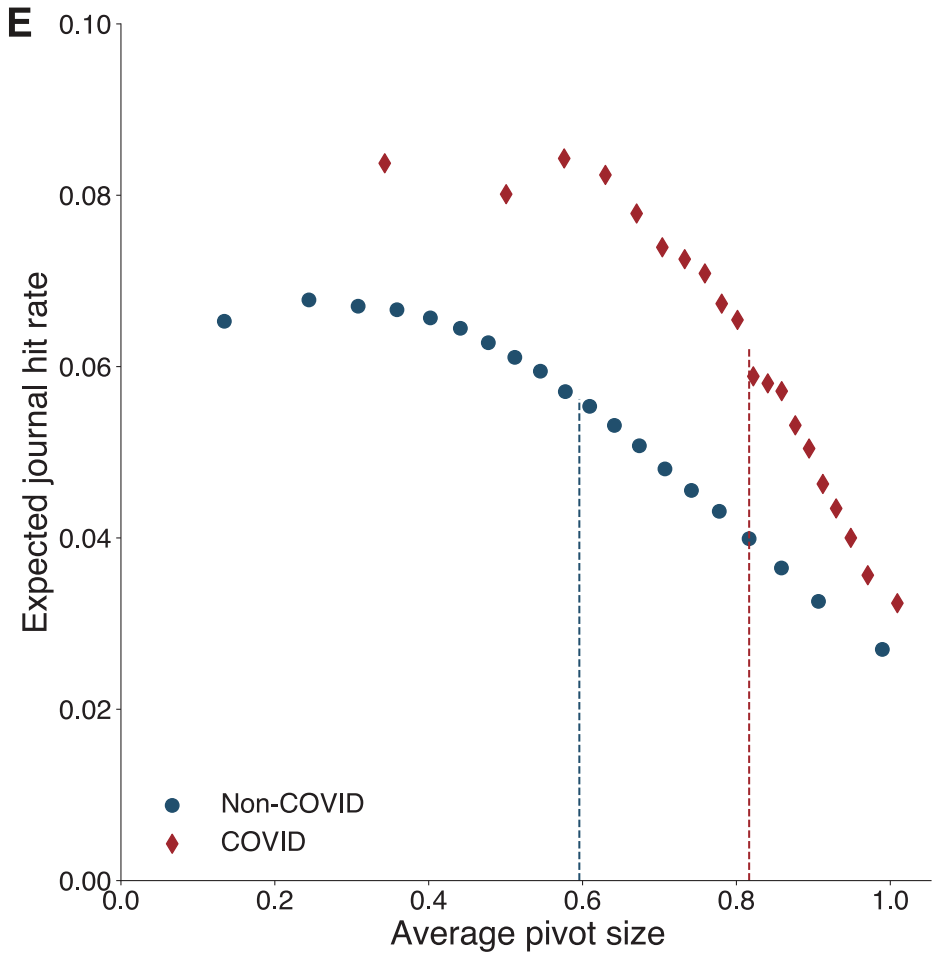
~7×

more likely that top-0.1% work is low-pivot than high-pivot

If large pivots were lottery tickets they would sometimes **win**. They do not.

COVID research earned a premium

and pivoting ate most of it

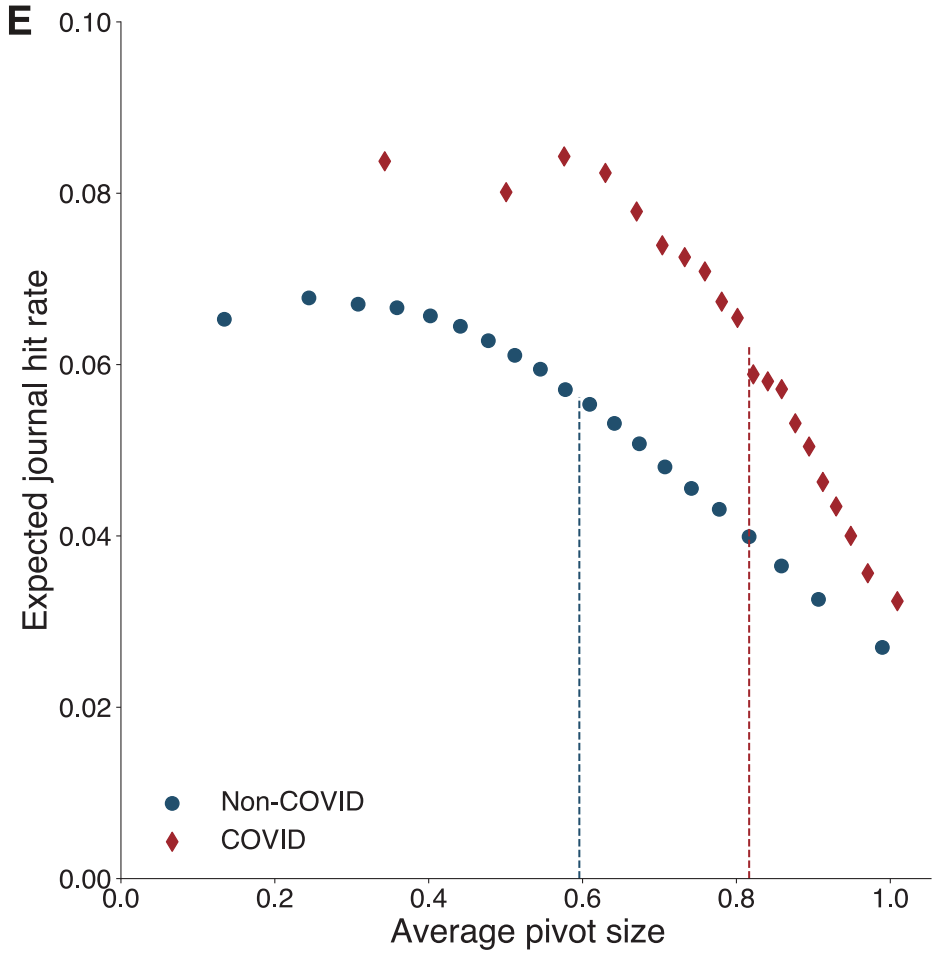


COVID work was **placed far better** than ordinary 2020 work at the same pivot size.

Yet the upper **45%** of COVID papers by pivot size did worse than non-COVID work at median pivot or below.

COVID research earned a premium

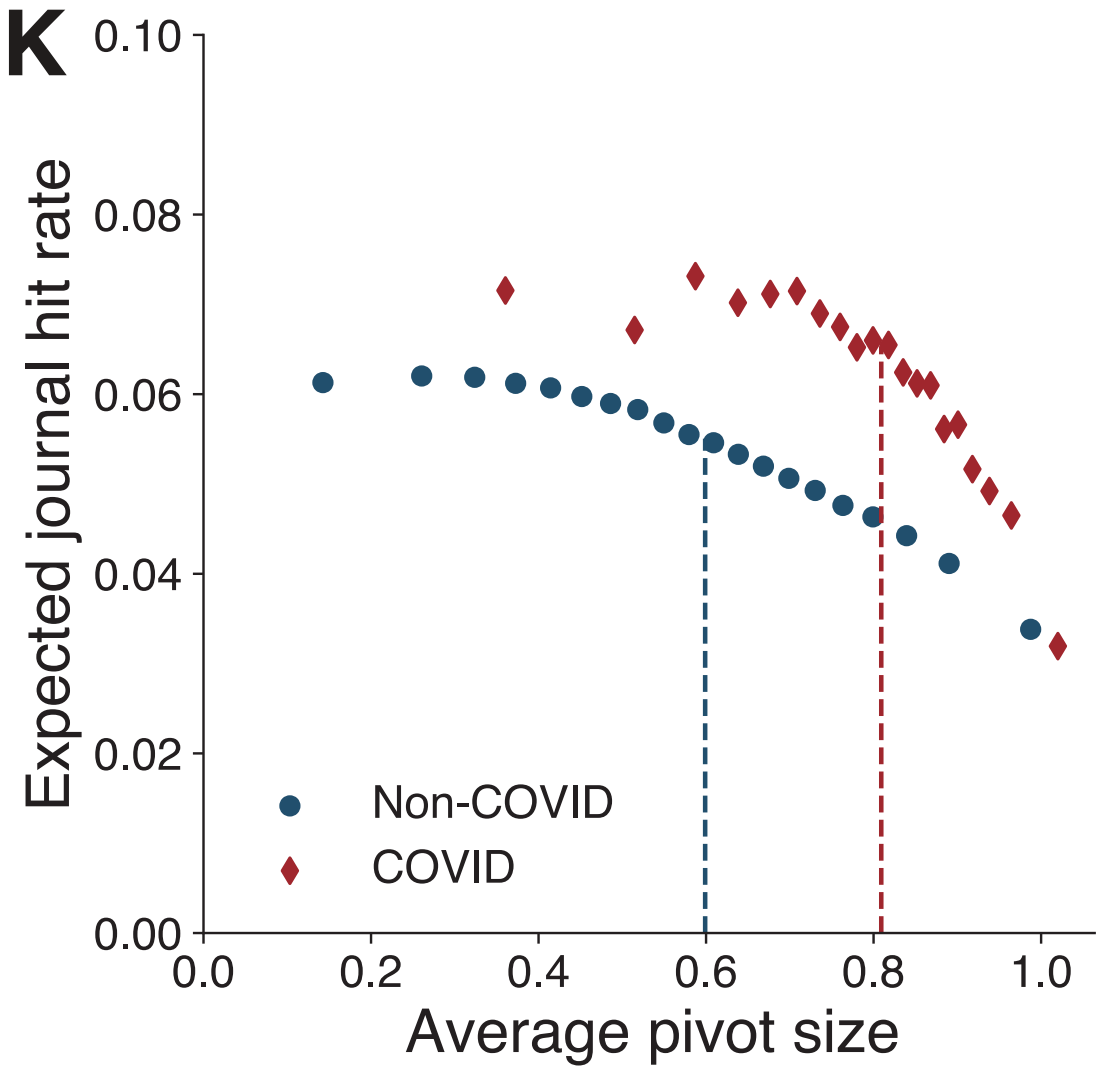
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Teams and funding do not rescue the pivot



COVID teams were **28% larger** and leaned unusually hard on new collaborators.

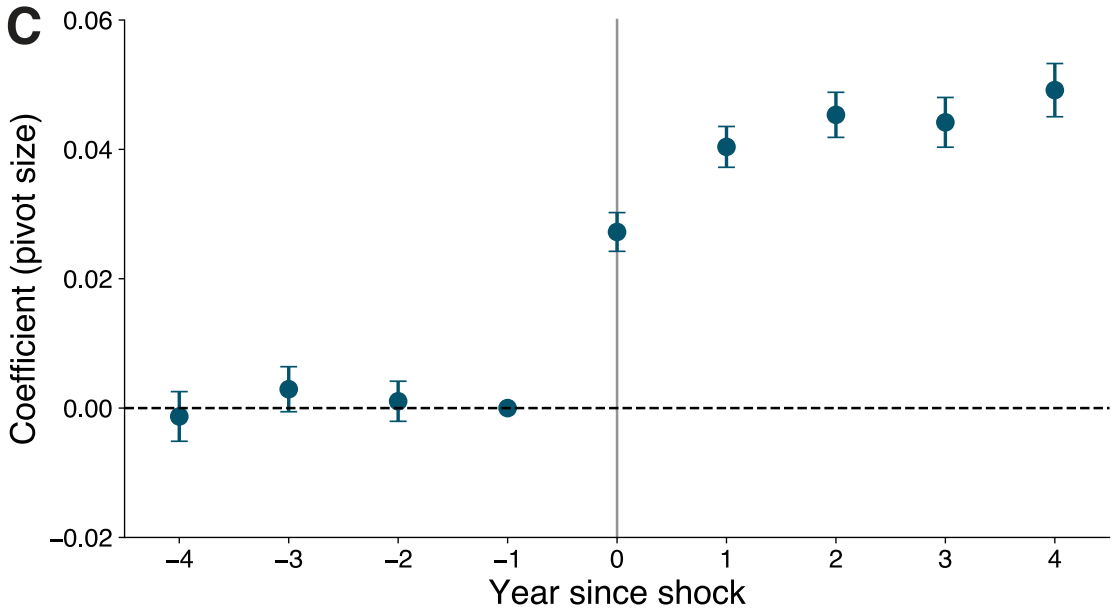
Net of career stage, productivity, team size, new coauthors and funding, the penalty **barely moves**.

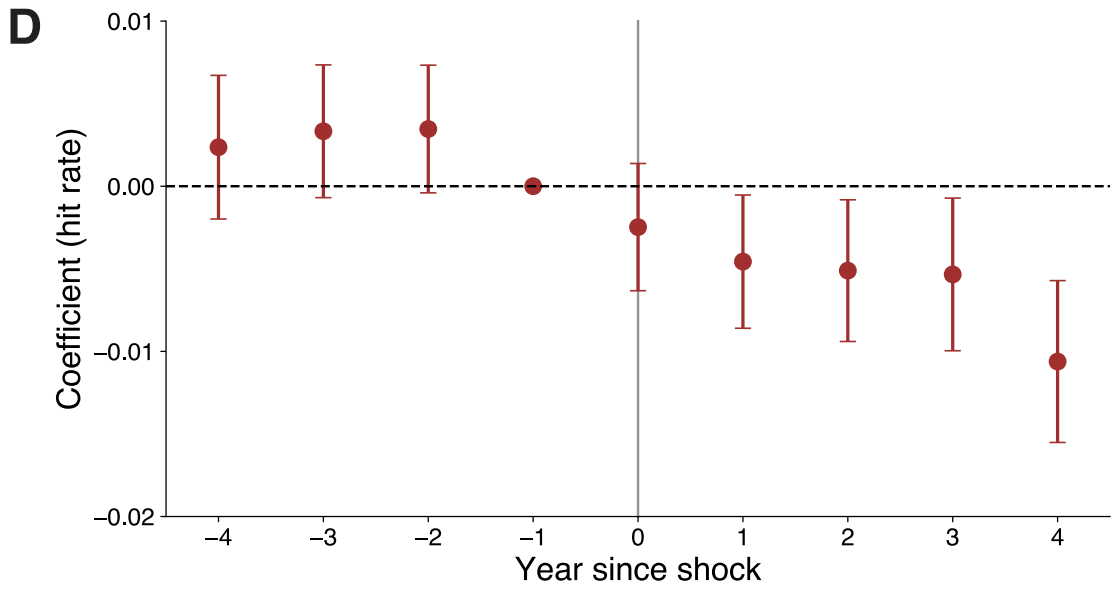
Adaptability has to be bought before you need it

You cannot pivot people on demand. You can only have already hired them — which makes a diverse portfolio of researchers a form of **risk management**, not a luxury.

> **“Chance favours only the prepared mind.”** — Pasteur

Yes — a retraction forces the same pattern





13,456 retractions. Treated authors pivot **+2.5 pp** more and lose **0.4 pp** of hit rate; for those citing the retracted work twice or more, **+3.7** and **–0.7**. All $p < 0.001$.

No — it is not just a change of audience

- Same researcher, same year, **same journal**: only 28% less steep
- It appears for distant evaluators — inventors and investors
- It holds at every career stage

Reputation explains **part** of it. Most survives holding the audience fixed.

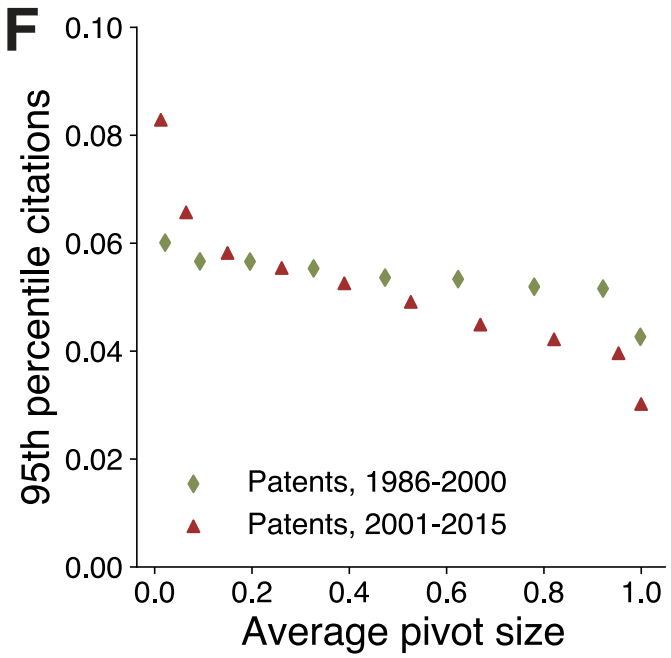
High-pivot work is novel but not conventional

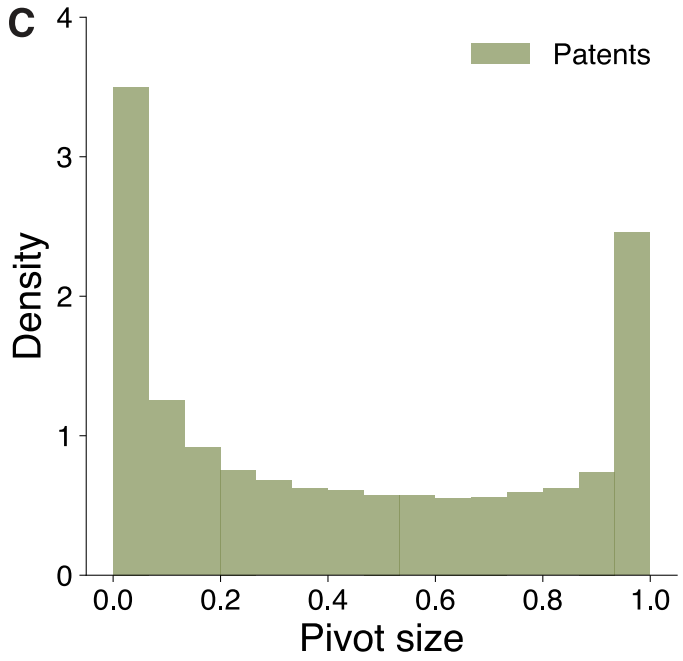
- Larger pivots bring **more** atypical combinations of prior knowledge
- Larger pivots bring distinctly **less** conventional grounding
- High-impact work has historically needed both at once

Pivoting researchers are equipped for novelty and **short on the conventional expertise** that usually carries it.

Inventors show the same steepening

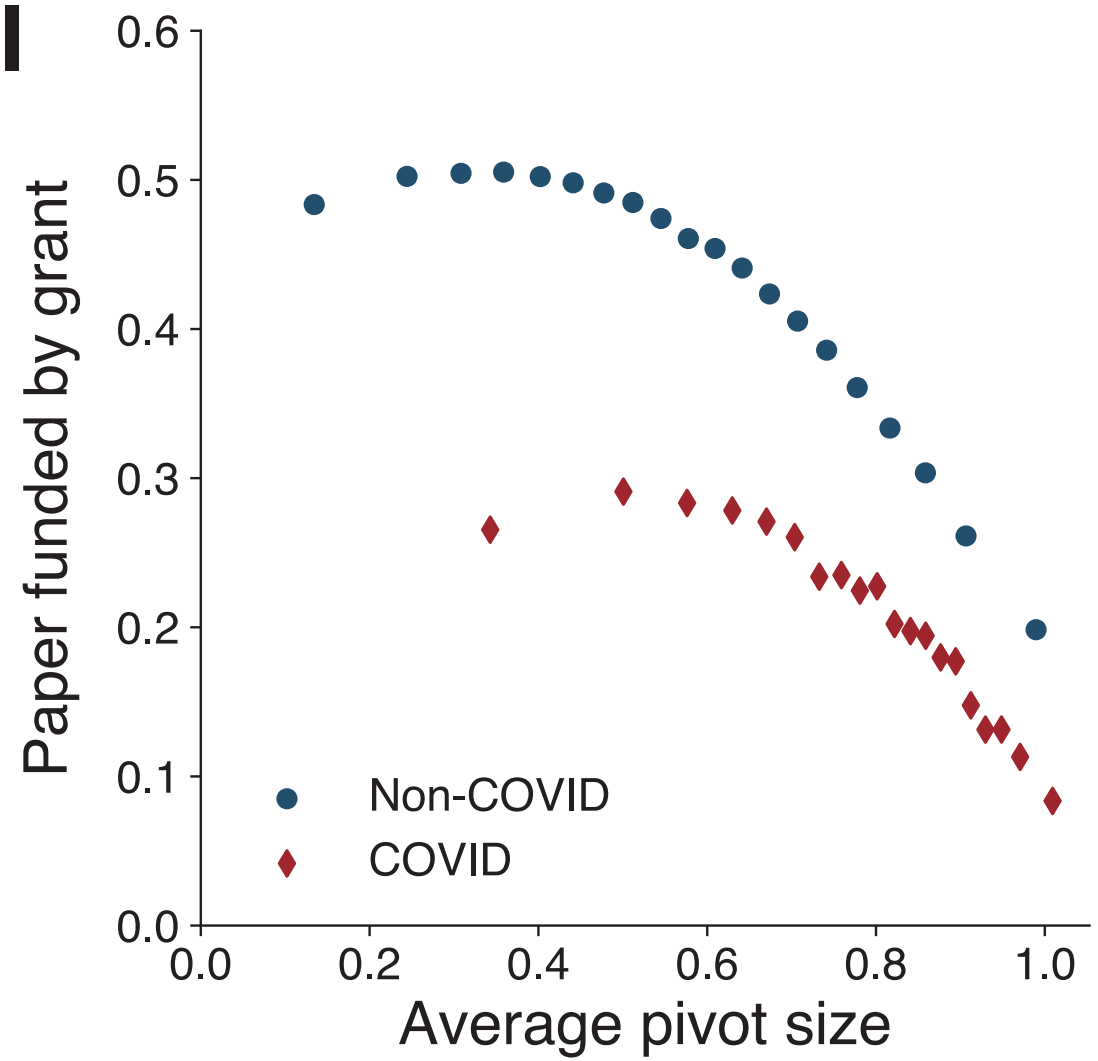
and a strikingly bimodal pivot distribution





Inventors either stay **very close** or jump **very far** — and the penalty holds in 91% of 127 classes.

Funding follows small pivots



Grant-supported work is concentrated at **small pivots**, and COVID work cited a grant less often at every pivot size.

Natural if grants fund **agendas**. The penalty appears in funded and unfunded work alike.